Roll No. | | | | | | | |

END SEMESTER EXAMINATION-2025

Name of the Course: B. Tech
Name of the Paper: **Automotive Materials and manufacturing Techniques**
Time: 3 Hours

Semester: IV
Paper Code: TME 412
Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(20marks)**	
(a)	Explain the primary mechanical and thermal properties that are essential when selecting materials for automotive engine blocks and structural components.	
(b)	Compare the advantages and limitations of ferrous and non-ferrous metals used in chassis and body frame components of passenger vehicles.	**CO 1/CO2**
(c)	Describe the use of composite materials in automotive body panels and explain how they contribute to weight reduction and corrosion resistance.	
Q2	**(20marks)**	
(a)	Compare die casting and gravity casting in terms of dimensional accuracy, cost-effectiveness, and their relevance in the automotive industry.	
(b)	Discuss the role of forging in producing high-strength automotive components such as crankshafts and how it improves fatigue resistance.	**CO 2**
(c)	Evaluate forming and machining techniques for manufacturing automobile body panels, focusing on surface finish and dimensional tolerances.	
Q3	**(20marks)**	
(a)	Explain the different types of welding used in automotive body assembly, such as MIG, TIG, and spot welding, along with their typical use cases.	
(b)	Describe the process parameters and metallurgical implications of friction stir welding in lightweight aluminum car body parts.	**CO 2 & CO 3**
(c)	Evaluate the need for hybrid joining techniques (adhesive + mechanical) in multi-material vehicle structures and crash safety improvement.	
Q4	**(20marks)**	
(a)	Describe the concept of Life Cycle Assessment (LCA) and its relevance in assessing material and energy use in automotive product development.	
(b)	Evaluate the role of recycling in automotive manufacturing and how closed-loop systems are being implemented by major OEMs.	**CO4/CO5**
(c)	Analyze the sustainability practices of any one automotive manufacturer and how they integrate robotics, green materials, and energy efficiency.	
Q5	**(20marks)**	
(a)	Present a case study on the use of carbon fiber composites in high-performance electric vehicles and discuss their impact on speed and efficiency.	
(b)	Review the development of electric vehicle (EV) battery housings and explain how material selection affects safety, cost, and thermal management.	**CO5/CO6**
(c)	Discuss the trends in autonomous vehicle material requirements, particularly sensor integration, lightweighting, and durability needs.	